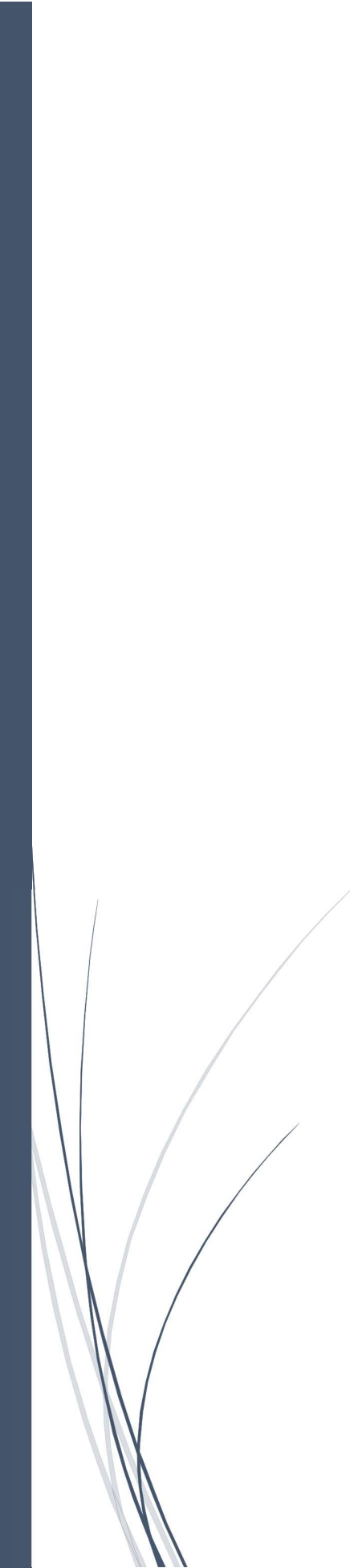




Voltage rail LC Filter Design

How to use the sheet

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VOLTAGE RAIL LC FILTER DESIGN

The equations used here is based on Webinar of Best PCB Design and Measurement Practices with Prof Eric Bogatin

Here is a detailed Excel sheet for automatic calculations

The Excel spreadsheet is titled "DC- Rail ferrite core+capacitor Calculator - Excel". It contains the following sections:

- INPUTS:**
 - Input Parameters: Value, Note
 - Used inductor (Ferrite of wire wound): 100.00, in μH
 - Used capacitor: 22.00, in μF
 - R-load: 3.30, in Ohm = $V_{\text{OUT}} / I_{\text{OUT}}$
- Automatic Equations calculation:**
 - R-DC of the inductor needed to make $Q=1$: 2.13, in Ohm
- Automatic Checklist of the selection:**
 - Does the R-DC of the inductor is appropriate to the load impedance: NO and will cause voltage drop
 - R-DC of the inductor will cause a significant voltage drop
- if YOU DON'T KNOW SOME PARAMETER OR TRYING VALUES "INPUTS":**
 - Capacitor used: 0.20, in μF
 - Inductance used: 4.70, in μH
 - R-DC Of the chosen inductor: 0.22, in Ohm*Given in datasheet*
 - V output: 5.00, To get R load assume the worst case "Highest current drawn"
 - I output: 10.00
- Automatic Equations calculation For the trying values section:**
 - R-load in Ohm: 0.50, R-DC of the inductor will cause a significant voltage drop
 - Q: 22.03, Over damped

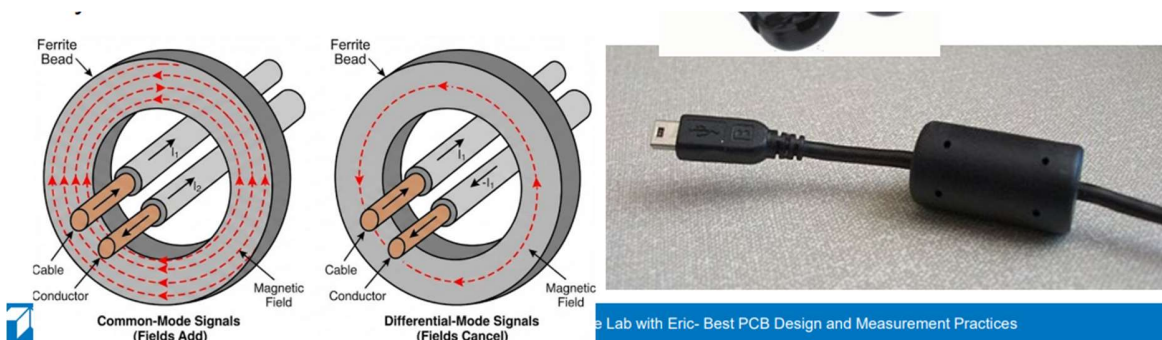
On the right side, there is an AC simulation circuit diagram showing a voltage source V_1 (1V, 1GHz) connected to a series combination of R_5 ($R=R_{\text{source}}$), R_8 ($R=R_{\text{L_DC}}$), L_3 ($L=L_{\text{L_L}}$), C_4 ($C=C_{\text{L_L}}$), R_9 ($R=ESR_{\text{C}}$), L_4 ($L=ESL_{\text{C_H}}$), C_3 ($C=C_{\text{C_F}}$), and R_{10} ($R=R_{\text{L_damping}}$). The output is taken across R_{10} . A note states: "If $R_{\text{source}} > R_{\text{DC}}$, source damps out ringing".

First Ferrite core is a high permeability material that increases the number of magnetic flux ring around inductor which 1A of current flows through which will lead to higher inductance for the common mode current path hence higher attenuations

It used only for attenuating the common mode current flows through it

Why?

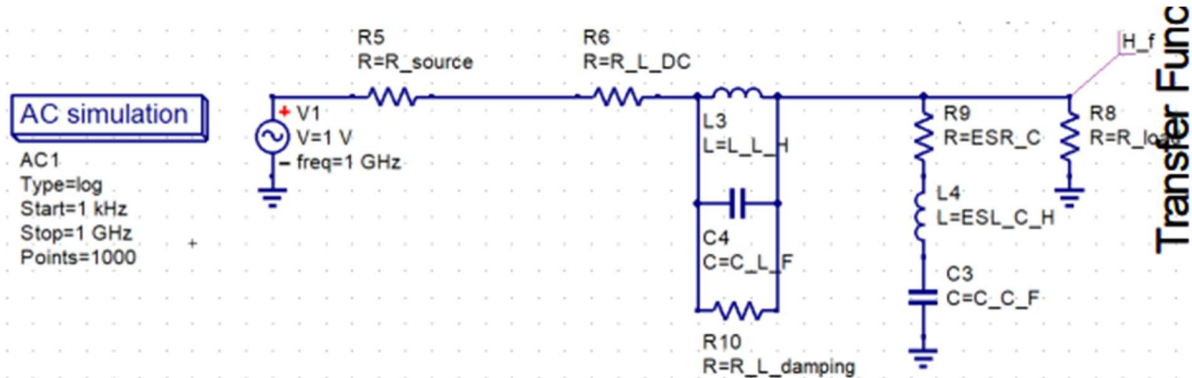
Because differential mode current is opposite in direction which will generates a magnetic flux that cancel each other



The approach here is based on damping the resonance that happens between the ferrite or the inductor and the capacitor forming the LC filter by Making ideally $Q=1$

VOLTAGE RAIL LC FILTER DESIGN

Any values other than $Q=1$ actually can make thing worse which can lead to oscillations and EMI Failure



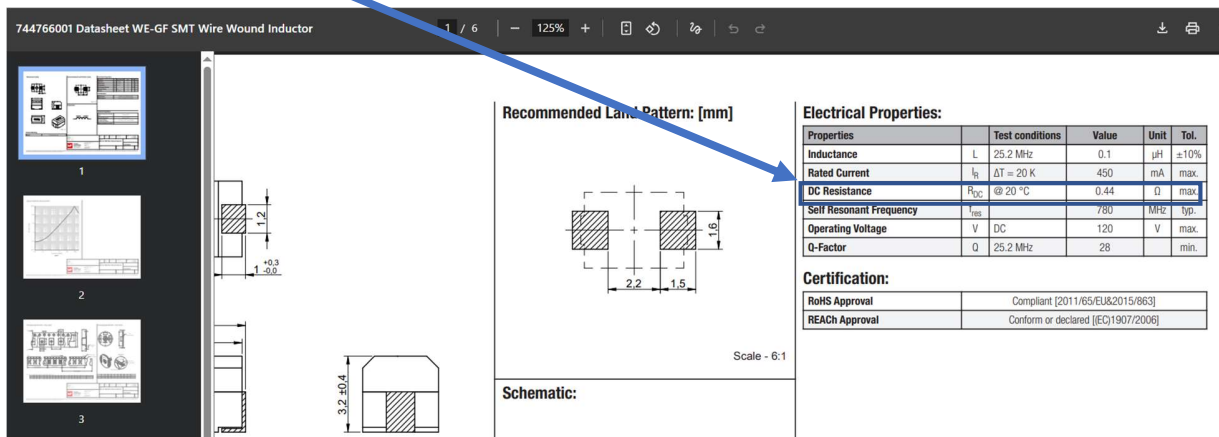
If $R_{\text{source}} > R_{\text{DC}}$, source damps out ringing

For $Q=1$, $R=\sqrt{L/C}$

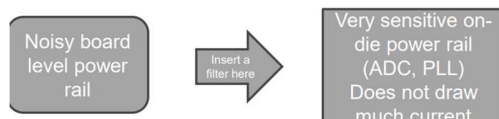
Where L is the inductance of the used inductor in μH

C is the capacitor used in μF

But R here is the R_{DC} of the inductor that maybe needed to be measured if not mentioned in datasheet of the inductor as shown



Note that you don't have to use L-C filter in all voltage rails only the critical one" VERY SENSITIVE for PDN NOISE Parts"



VOLTAGE RAIL LC FILTER DESIGN

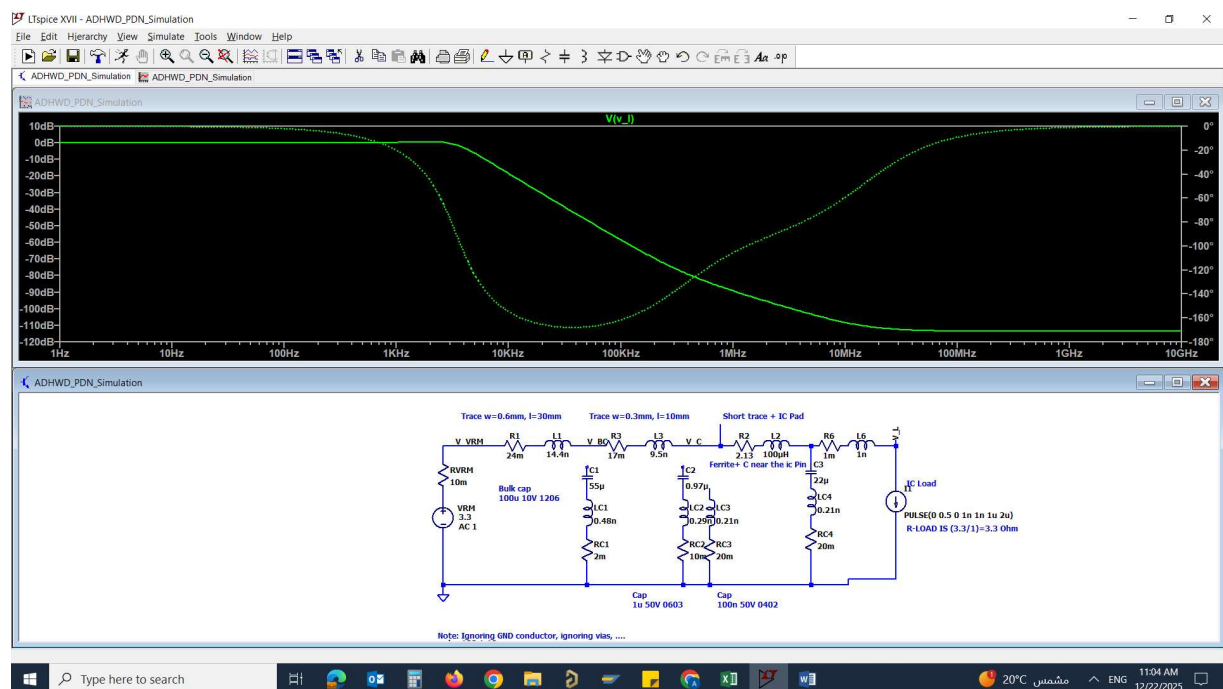
The peak here is formed by the parasitic Inductance of the PDN and the decoupling capacitor placed near the IC

Let's use the sheet to figure out the L-C values

I have used a random value for R-load and c and L just prove of concept nothing else

INPUTS		
Input Parameters	Value	Note
Used inductor (Ferrite or wire wond)	100.00	In μH
Used capacitor	22.00	In μF
R-load	3.30	in Ohm = $V_{\text{OUT}}/I_{\text{OUT}}$
Automatic Equations calaculation		
R-DC of the inductor needed to make $Q=1$	2.13	in Ohm
Automatic Checklist of the selection		
Does the R-DC of the inductor is appropriate to the load impedance	NO and will cause voltage drop	R-DC of the inductor will cause a signifecnt voltage drop

Let's put these values and see the waveform at the load which is the vcc rail of the IC



LOOK IT IS Damped but wait it very bad wave it suffers from loss after some low frequency

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Lets see the sheet again

INPUTS		
Input Parameters	Value	Note
Used inductor (Ferrite of wire wond)	100.00	In μH
Used capacitor	22.00	In μF
R-load	3.30	in Ohm = $V_{\text{OUT}}/I_{\text{OUT}}$
Automatic Equations calaculation		
R-DC of the inductor needed to make $Q=1$	2.13	in Ohm
Automatic Checklist of the selection		
Does the R-DC of the inductor is appropriate to the load impedance	NO and will cause voltage drop	R-DC of the inductor will cause a signifecnt voltage drop

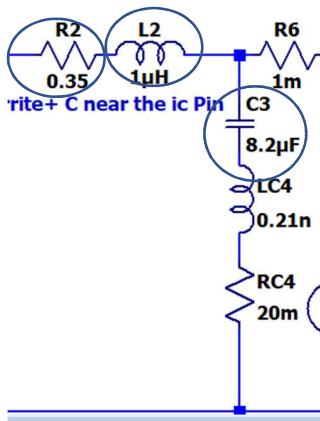
Now we see why because of the R-DC having a voltage drop because the output load is small

Also note the automatic checklist explaining this point

You have to choose another inductor which have at least $R\text{-DC} \cdot 10 <$ the load Resistance which is the ic

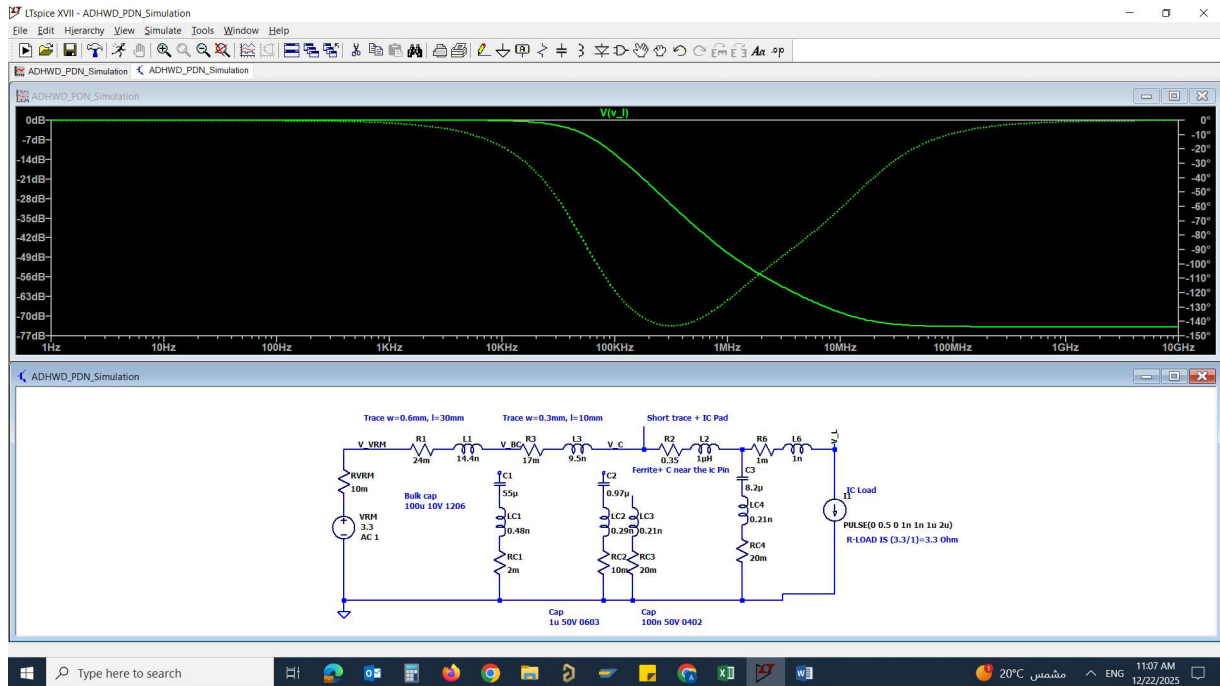
Let use another random L-C

INPUTS		
Input Parameters	Value	Note
Used inductor (Ferrite of wire wond)	1.00	In μH
Used capacitor	8.20	In μF
R-load	3.30	in Ohm = $V_{\text{OUT}}/I_{\text{OUT}}$
Automatic Equations calaculation		
R-DC of the inductor needed to make $Q=1$	0.35	in Ohm
Automatic Checklist of the selection		
Does the R-DC of the inductor is appropriate to the load impedance	NO and will cause voltage drop	R-DC of the inductor will cause a signifecnt voltage drop



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HERE IS THE Waveform of the VCC Rail of the IC (Load)



Done

Note: This Way was proven in the webinar with real time measurement by demo board.